

Sheet 3 - optimisation

1. $Df(x,y) = (4x^3 + 2x - 6y \quad -6x + 6y)$

which $= 0$ at critical points (defn of f.o.c)

$$\Rightarrow x=y, \Rightarrow 4x^3 - 4x = 0$$

$$= 4x(x^2 - 1) = 0$$

$$0 \quad \pm 1$$

So the critical points are $(0,0), (1,1), (-1,-1)$.

Now consider the Hessian $D^2f = \begin{pmatrix} 12x^2 + 2 & -6 \\ -6 & 6 \end{pmatrix}$

with $|D^2f| = 72x - 24$.

At $(0,0)$, the leading principal minors are 2, -24.

$\hookrightarrow D^2f$ indefinite $\Rightarrow$ saddle point

At $(1,1)$ they're 14, 48 $\Rightarrow D^2f$ is p.d., so strict local min

At $(-1,-1)$ they're 14, -48 $\Rightarrow D^2f$ is ind., saddle point.

2. $Df(x,y) = (2x + y + 3 \quad 4y + x + 19)$

$$Df^2(x,y) = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix} =: H.$$

$|H| = 7$, $\text{tr}(H) = 6$. So $\sum \lambda_i, \prod \lambda_i > 0$ i.e. the eigenvalues are all strictly positive, for any (x,y) .

By eigenvalue test, H is ~~strictly~~ positive definite, for any (x,y) .
So f is strictly convex.

Using FOCs, $Df = 0$, the single stationary point is
at $z_y = -35 \Rightarrow y = -5, x = 1$.

And since f is strictly convex, $(1, -5)$ is a global minimum.

3.a) $\max_{x,y} x^2 + y^2 \quad \text{s.t.} \quad x^2 + xy + y^2 = 3.$

The Lagrangian $L(x,y;\lambda) = x^2 + y^2 - \lambda(x^2 + xy + y^2 - 3).$

Then $D_x L(x,y;\lambda) = Df(x,y) - \lambda Dh(x,y)$ where $h(x,y) = x^2 + xy + y^2 - 3$

$= (2x - \lambda(2x+y) \quad 2y - \lambda(x+2y)) = 0$ at each

* at all

feasible points, including - in particular - optima.

constrained optimum ~~announces~~ by Lagrange theorem, ~~requires that~~ since $Dh = (2x+y \quad x+2y) = 0$ only at $(0,0)$ which is not feasible, so it's full rank* (constraint).

~~So $2x - \lambda(2x+y) = 0$ and $2y - \lambda(x+2y) = 0$~~

Ans. Subtracting FOCs for x, y :

$2(x-y) - \lambda(x-y) = 0 \Rightarrow (\lambda-2)(x-y) = 0.$

So $\lambda = 2$ or $x = y$.

(i) ~~$2x - 4x - 2y = 0 \Rightarrow x = -y$~~

From the constraint, then

$x^2 - x^2 + x^2 = 3 \Rightarrow x = \pm\sqrt{3}$

So solutions $(\sqrt{3}, -\sqrt{3})$ and $(-\sqrt{3}, \sqrt{3})$ with objective 6

(ii) From the constraint, then

$x^2 + x^2 + x^2 = 3 \Rightarrow x = \pm 1$

So solutions $(1, 1)$ and $(-1, -1)$ with objective 2

Since our feasible region is a compact set, we could use Weierstrass and directly conclude that ^{global} minima and maxima exist on S (and by Lagrangian necessity must be among the above candidates). But we'll check SOC's to be sure.

Q3 all correct, well done!

The bordered Hessian is
$$\begin{pmatrix} 0 & D_h(x^*) \\ D_h(x^*)^T & D^2_x L(x^*; \lambda^*) \end{pmatrix}$$

$$\begin{pmatrix} 0 & 2x+y & x+2y \\ 2x+y & 2-2\lambda & -\lambda \\ x+2y & -\lambda & 2-2\lambda \end{pmatrix} \quad \text{where } x^* = \sqrt{3}$$

We need to check the sign of $|\bar{H}|$.

At solutions (i), doing Laplace expansion along first row, we have

$$|\bar{H}| = 0 \cdot 1 \cdot 1 - (2\sqrt{3} - \sqrt{3}) \cdot 1 \cdot 1 + \sqrt{3} \cdot 1 \cdot 1$$

$$= -\sqrt{3} \cdot (\sqrt{3} - 4 + \lambda \cdot -\sqrt{3})$$

$$= -\sqrt{3} \cdot (\sqrt{3} - \lambda - 4 \cdot -\sqrt{3}) = 24 > 0 \quad \text{so max at } (\sqrt{3}, -\sqrt{3}) \text{ and similarly } (-\sqrt{3}, \sqrt{3})$$

At solutions (ii), we have $\lambda^* = 2/3$. So $|\bar{H}|$

$$|\bar{H}| = 0 \cdot 1 \cdot 1 + 4 \cdot \begin{vmatrix} 3 & -2/3 \\ 3 & 2/3 \end{vmatrix} + 3 \cdot \begin{vmatrix} 3 & -2/3 \\ 3 & -2/3 \end{vmatrix}$$

$$= -3 \cdot (6/3 + 6/3) + 3 \cdot (-6/3 - 6/3) = -24 < 0$$

so min at $(1, 1)$ and similarly $(-1, -1)$.

4. i) BC must bind, because utility is strictly increasing in consumption. So leaving income unspent cannot produce a maximum, as they could feasibly consume more of either good.

Consumption of both goods must be strictly positive because these are Cobb-Douglas preferences and which we can write equivalently as $v = 2 \ln x + \ln y$ ($\ln$ is monotonic increasing) and if $x, y = 0$ then their marginal utility of that good $= \infty$. But then it's not optimal to consume none of it but some of the other.
 (Or, note that if x or $y = 0$, then utility is 0, but strictly positive utility is attainable, e.g. at $(1, 1)$.)

ii) $L(x, y; \lambda) = 2 \log x + \log y - \lambda(2x + 3y - 9)$

We know from (i) we have an interior solution with BC binding, so we can ignore the nonnegativity constraints as they don't bind.

The FOCs are $\frac{\partial L}{\partial x} = 2/x - 2\lambda \stackrel{!}{=} 0$

$\frac{\partial L}{\partial \lambda} = 2x + 3y - 9 \stackrel{!}{=} 0$

$\frac{\partial L}{\partial y} = 1/y - 3\lambda \stackrel{!}{=} 0$

which are solved by $y^* = \frac{1}{3} \cdot \frac{9}{3} = 1$ } (Since Cobb-Douglas,
 $\lambda^* = \frac{1}{3}$ } or noting $\frac{1}{3}y = \lambda = \frac{1}{2}x \Rightarrow x = 3y$
 $x^* = \frac{2}{3} \cdot \frac{9}{2} = 3$

iii) We need to construct $\bar{H}$ and check signs of leading principal minors.

$$\bar{H} = \begin{pmatrix} 0 & -2 & -3 \\ -2 & -2x^{-2} & 0 \\ 3 & 0 & -y^{-2} \end{pmatrix}$$

For 2 variables and 2 constraint, a strict local maximum requires $|\bar{H}| > 0$. Evaluating at (3, 1),

$$|\bar{H}| = 0 \times 1 \times 1 - 2 \times \begin{vmatrix} 2 & 0 \\ 3 & -1 \end{vmatrix} + 3 \times \begin{vmatrix} 2 & -2 \\ 3 & 0 \end{vmatrix}$$

$$= 0 + 2 \times 2 + 3 \times 6/9 = 4 > 0$$

So indeed a strict local max.

and u is continuous.

iv) The feasible region is a convex, bounded set. So by Weierstrass, it attains a ~~unique~~ global maximum. And we found the sole stationary point at (1, 3) is a strict local maximum. So it's the unique global maximum.

5. the firm solves the cost-minimisation problem:

$$\begin{aligned} \min_{K, L} \quad & rK + wL \quad \text{s.t.} \quad (K+1)^{1-\alpha} (L+1)^\alpha \geq y+1 \\ & \text{with } \alpha \in (0, 1) \\ & \text{and } K \geq 0, L \geq 0 \end{aligned}$$

Note that we know the production constraint binds at a minimum. If $Y(K, L) > y$, the firm could reduce either K or L (in which costs are strictly increasing) and still be at a feasible point.

Take the Lagrangian $\mathcal{L}(K, L, \lambda) = rK + wL - \lambda((K+1)^{1-\alpha} \cdot (L+1)^\alpha - y+1)$

$$\text{with FOCs } \frac{\partial \mathcal{L}}{\partial K} = r - (1-\alpha) \lambda (K+1)^{-\alpha} \cdot (L+1)^\alpha \stackrel{!}{=} 0$$

$$\frac{\partial \mathcal{L}}{\partial L} = w - \alpha \lambda (K+1)^{1-\alpha} \cdot (L+1)^{\alpha-1} \stackrel{!}{=} 0$$

$$\text{and } \frac{\partial \mathcal{L}}{\partial \lambda} \stackrel{!}{=} 0 \text{ i.e. production constraint binding.}$$

Here, we've assumed the positivity constraints don't bind, i.e. at an interior solution; we'll analyse corner solutions shortly.

Rearranging for r and w and dividing, we have tangency condition:

$$\frac{L}{w} = \frac{(1-\alpha)(K+1)^{-\alpha}}{\alpha(K+1)^{1-\alpha}}$$

which is always satisfied, since $\alpha < 1$, K and L

$$\text{So } K+1 = \frac{(1-\alpha)(L+1)}{\alpha} \cdot \frac{w}{r};$$

$$\therefore \left(\frac{1-\alpha}{\alpha} \cdot \frac{w}{r} \right)^{1-\alpha} \cdot (L+1) = y+1 \Rightarrow L+1 = \left(\frac{1-\alpha}{\alpha} \cdot \frac{w}{r} \right)^{\alpha-1} \cdot (y+1)$$

$$\text{and } K+1 = \left(\frac{1-\alpha}{\alpha} \cdot \frac{w}{r} \right)^{\alpha} \cdot (y+1)$$

$$\text{With cost } r \cdot \left(\left(\frac{1-\alpha}{\alpha} \cdot \frac{w}{r} \right)^{\alpha} \cdot (y+1) - 1 \right) + w \cdot \left(\left(\frac{1-\alpha}{\alpha} \cdot \frac{w}{r} \right)^{\alpha-1} \cdot (y+1) - 1 \right)$$

This solution is valid when $K, L \geq 0$. So, we require

$$L: \left(\frac{1-\alpha}{\alpha} \cdot \frac{w}{r} \right)^{\alpha-1} \cdot (y+1) \geq 1, \quad K: \left(\frac{1-\alpha}{\alpha} \cdot \frac{w}{r} \right)^{\alpha} \cdot (y+1) \geq 1.$$

~~we have a constraint~~ When wages are high relative to rent, we might get an implied optimum of L which is negative, and the converse for K .

~~At the optimal point of $L \geq 0, K \geq 0$~~ After arranging the above, we have:

$$\left(\frac{1}{y+1} \right)^{1/\alpha} \cdot \frac{\alpha}{\alpha-1} \leq \frac{w}{r} \leq \left(\frac{1}{y+1} \right)^{1/\alpha-1} \cdot \frac{\alpha}{1-\alpha}$$

$\rightarrow$ ineq. sign flips due to $1/\alpha-1 < 0$

Or for $1/w$, take reciprocals of both sides and flip $\leq$ to $\geq$:

$$(y+1)^{\frac{1}{\alpha-1}} \cdot \frac{1-\alpha}{\alpha} \leq \frac{r}{w} \leq (y+1)^{\frac{1}{\alpha}} \cdot \frac{\alpha-1}{\alpha}$$

If the L inequality fails, i.e. rent is very cheap compared to wages, the constrained optimum will be to purchase $L=0$, and

$K = (y+1)^{1/\alpha} - 1$. Similarly if K inequality fails, purchase $K=0$, $L = (y+1)^{1/\alpha} - 1$, with corresponding costs in terms of K, L, w .

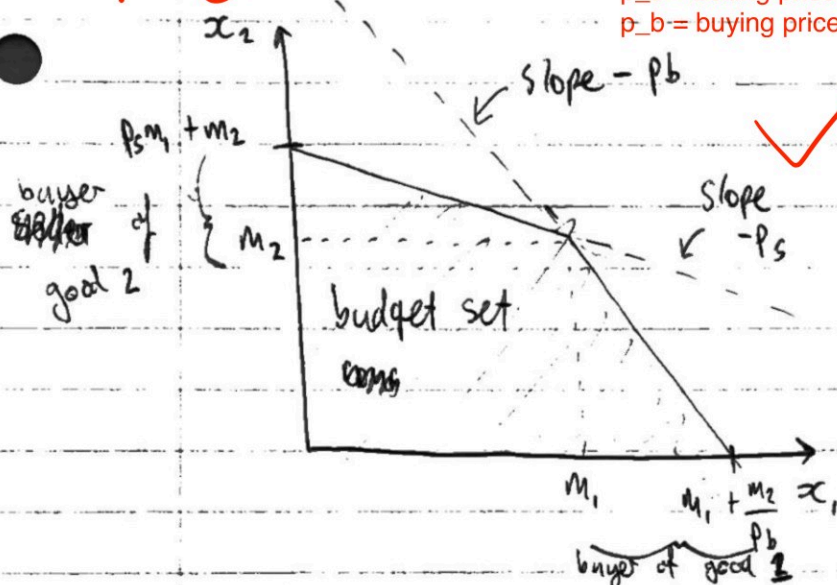
Because this is a convex problem, the FOCs are sufficient for global min.

6. $u(x_1, x_2) = x_1^\beta + x_2^\beta$, $0 < \beta < 1$. $x_1 \geq 0$, $x_2 \geq 0$

$$p_s x_1 + x_2 \leq p_s m_1 + m_2, \quad p_b x_1 + x_2 \leq p_b m_1 + m_2, \\ 0 < p_s < p_b.$$

There's a bid-ask spread: the price the household is able to sell their endowment at is worse (lower) than what they need to buy at. p_s is the number of units of good 2 ^{sold} that good 1 sells for; p_b is the number of units of good 2 you could buy ^{instead of} 1 unit of good 1.

p_s = selling price of good 1
 p_b = buying price of good 1



a) $x_2 \leq -p_s x_1 + p_s m_1 + m_2$

b) $x_2 \leq -p_b x_1 + p_b m_1 + m_2$
↑ steeper

intersects: $x_1(p_b + p_s) = m_1(p_b + p_s)$
i.e. at initial endowment.

We have two equations describing the budget set because if they're a net seller of good 1 then their ^{rate of conversion} is worse (in terms of acquiring ^{marginally} more of good 2) than if they were a net buyer, due to the price spread. ^{Terms of trade}

There are three relevant cases to consider:

- Corner solution (no trade, remains at initial endowment) This solution will not be in a corner but the middle
- Net seller of good 1 (tangency of IC with upper budget constraint)
- Net buyer of good 1 (IC tangent to lower budget constraint)

We know the only corner solution where constraints bind with $=$ is at the initial endowment rather than x_1 or $x_2 = 0$, because at those points MU of the not-consumed good $\rightarrow \infty$ which means it can't be optimal to buy a finite amount of other good given finite price ratio (Endow conditions).

- Given that the utility function to be maximised is concave and all $\leq$ constraints linear, this is a concave problem. So we can find global optima just by identifying local maxima.
 - At interior solutions, tangency condition is $MRS = -p_j$ for the relevant $j \in \{3, b\}$.
- ~~At the seller of good 1, tangency condition is that~~
- $$MRS : -\frac{\partial u}{\partial x_1} \div \frac{\partial u}{\partial x_2} = -\left(x_1^{\beta-1} / x_2^{\beta-1}\right) = -\left(\frac{x_1}{x_2}\right)^{\beta-1}$$

(ii) Seller of good 1: $j=3$, so $x_1 = x_2 \cdot p_3^{\frac{1}{\beta-1}}$;

$$p_3 x_1 + x_2 = p_3 m_1 + m_2$$

$$\therefore x_2 = p_3 (m_1 - x_1) + m_2$$

$$\Rightarrow x_1 = \frac{p_3 m_1 + m_2}{p_3 \cdot p_3^{\frac{1}{\beta-1}} + 1} \cdot p_3^{\frac{1}{\beta-1}} = \frac{p_3 m_1 + m_2}{p_3 + p_3^{\frac{1}{\beta-1}}}$$

What will x_2 be?

We require $x_1 < m_1$ for consistency of being a seller.

$$\text{So } p_3 m_1 + m_2 < p_3 m_1 + m_1 p_3^{\frac{1}{\beta-1}} \Rightarrow \left(\frac{m_2}{m_1}\right)^{1-\beta} < p_3$$

i.e. the absolute value of the MRS evaluated at initial endowment, $\left(\frac{m_1}{m_2}\right)^{1-\beta}$ must be strictly less than the sale price p_3 for this solution to be valid.

Similarly for (iii) Buyer of good 1, we have

$$\text{Solution } x_2 = p_b (m_1 - x_1) + m_2 \text{ with } x_1 > m_1$$

$$\Rightarrow x_1 = \frac{p_b m_1 + m_2}{p_b + p_b^{\frac{1}{\beta-1}}} \text{ which is valid when } |MRS|_{\omega} \geq p_b$$

The corner solution will be where one $x = 0$; you should think why such a "corner solution" won't exist in this case

If $p_s \leq |MRS|_{\omega} \leq p_b$, then (i) Corner solution (autarky) will be optimal, with both constraints binding; household consumes (m_1, m_2) .

What is this quantity?

Check solutions to learn how to do with problem by setting up a Lagrangian! It's pretty cool

Intuitively: the preferences exhibit diminishing marginal utility. If even at the initial endowment the consumer doesn't derive enough utility from buying units of one good to compensate for the spread with selling the other, it'll never be worth it for them to trade at all. Nonzero transaction costs make the budget set smaller and mean fewer trades occur.